

# Output Analysis, Verification, and Validation

CSS142P Modelling and Simulation  
Week 7 | Can we trust the answer?



# A simulation always gives you a number. This week is about whether anyone should believe it.

- Last week, ten runs of the same canteen model gave average waits from 0.88 to 6.91 minutes.
- A manager will spend real money on the number you report.
- So every result needs three answers: **How precise is it? Is the code right? Is the model right?**
- Output analysis answers the first. Verification and validation answer the other two.

# What this week covers



## SECTION 1

### From one run to an honest answer

Replications, the sample mean, and the confidence interval, worked by hand.



## SECTION 2

### How many runs, and what kind of study

Choosing precision, and telling terminating studies from steady-state ones.



## SECTION 3

### Verification

Did we build the model right?  
Checks that catch silent bugs.



## SECTION 4

### Validation

Did we build the right model?  
Comparing with the real system.



## SECTION 5

### Sensitivity and robust advice

What if our inputs are a little wrong? Reporting results people can trust.

## LEARNING OUTCOMES

# What you will be able to do

By the end of this week, you should be able to:

- 1 Explain why a single simulation run is not a result.
- 2 Build a 95 percent confidence interval from replications, step by step.
- 3 Work out how many replications a required precision needs.
- 4 Tell a terminating study from a steady-state study, and treat each correctly.
- 5 Verify a model with conservation and sanity checks, and validate it against real data.
- 6 Test how sensitive a recommendation is to uncertain inputs.

## SECTION 1

## 01

# From One Run to an Honest Answer

*One day is a story. Ten days are evidence.*

## IN THIS SECTION

- Why one run misleads
- Replications
- The sample mean
- Standard error
- Confidence intervals
- Reading an interval

## SECTION 1: ONE RUN TO AN ANSWER

# Why one run misleads

- **Random output:** when a model uses random inputs, its output is random too.
- Each seed gives one possible day at the canteen.
- Ten seeds gave average waits from **0.88** to **6.91** minutes.
- Report the lowest day and the canteen looks fine. Report the highest and it looks broken.
- Both are real outcomes of the same model. Neither one alone is the answer.



## EVERYDAY EXAMPLE

## Judging a restaurant by one visit

One slow dinner does not mean the restaurant is slow. You would want several visits before you told a friend. A simulation needs several runs for the same reason.

## SECTION 1: ONE RUN TO AN ANSWER

# Replications and the sample mean

- **Replication:** one independent run of the model, with its own random seed.
- Each replication gives ONE number, such as the average wait that day.
- **Sample mean:** add the replication results and divide by how many there are.
- The sample mean is our best single estimate of the true average.
- More replications give a more trustworthy mean, but never a perfect one.



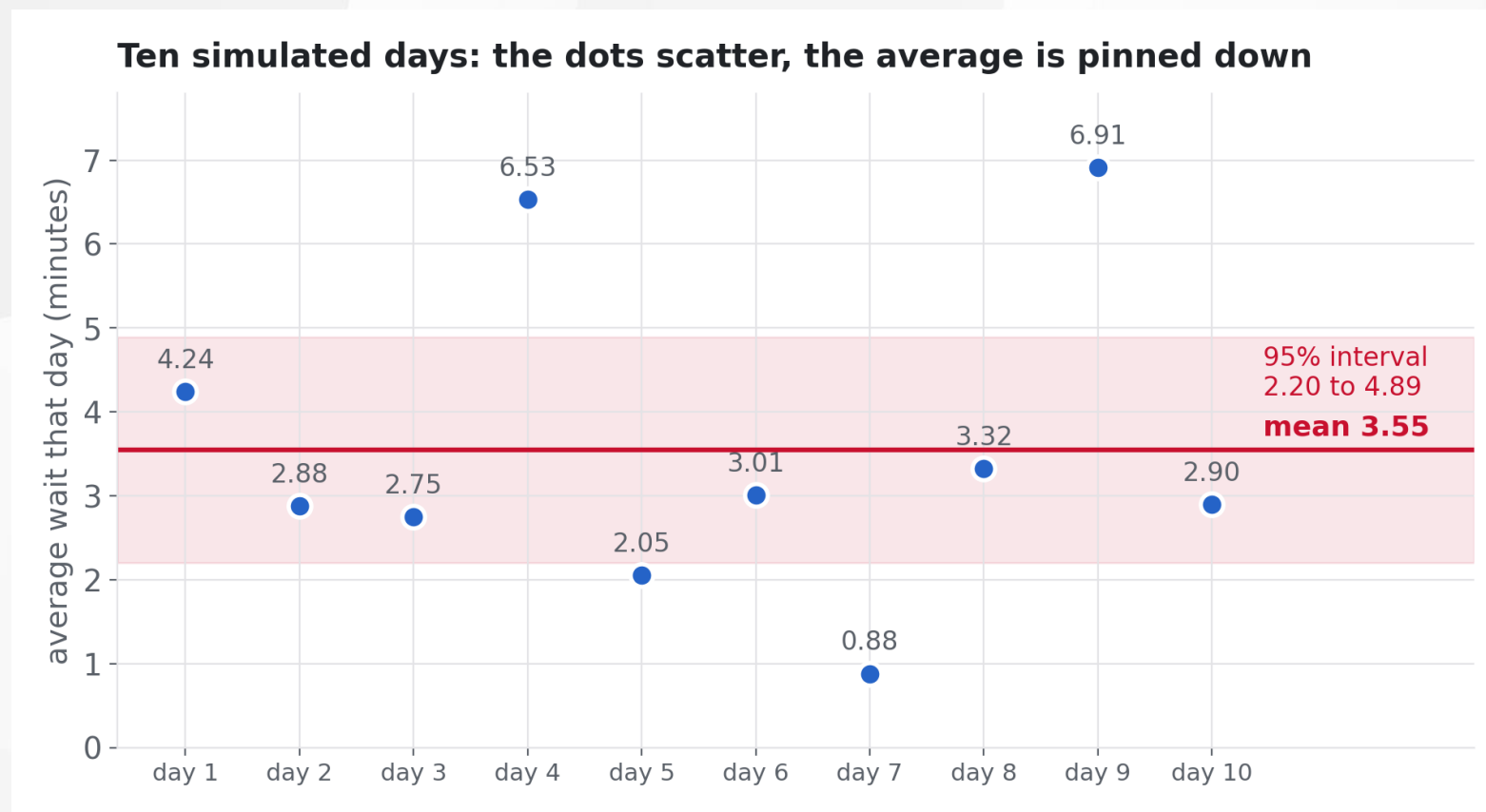
## QUICK CALCULATION

**Our ten days**

$4.24 + 2.88 + 2.75 + 6.53 + 2.05$   
 $+ 3.01 + 0.88 + 3.32 + 6.91 +$   
 $2.90 = 35.46$ . Divide by 10: the  
sample mean is 3.55 minutes.

## SECTION 1: ONE RUN TO AN ANSWER

# Ten days, one pinned-down average



*Individual days scatter widely. The average of ten days is known far more precisely.*

# Spread of the days versus uncertainty of the average

## 1.88

### Standard deviation

How much one day differs from another. Days really do vary this much.

## 0.60

### Standard error

How unsure we are about the AVERAGE. It is the standard deviation divided by the square root of 10.

## 1.35

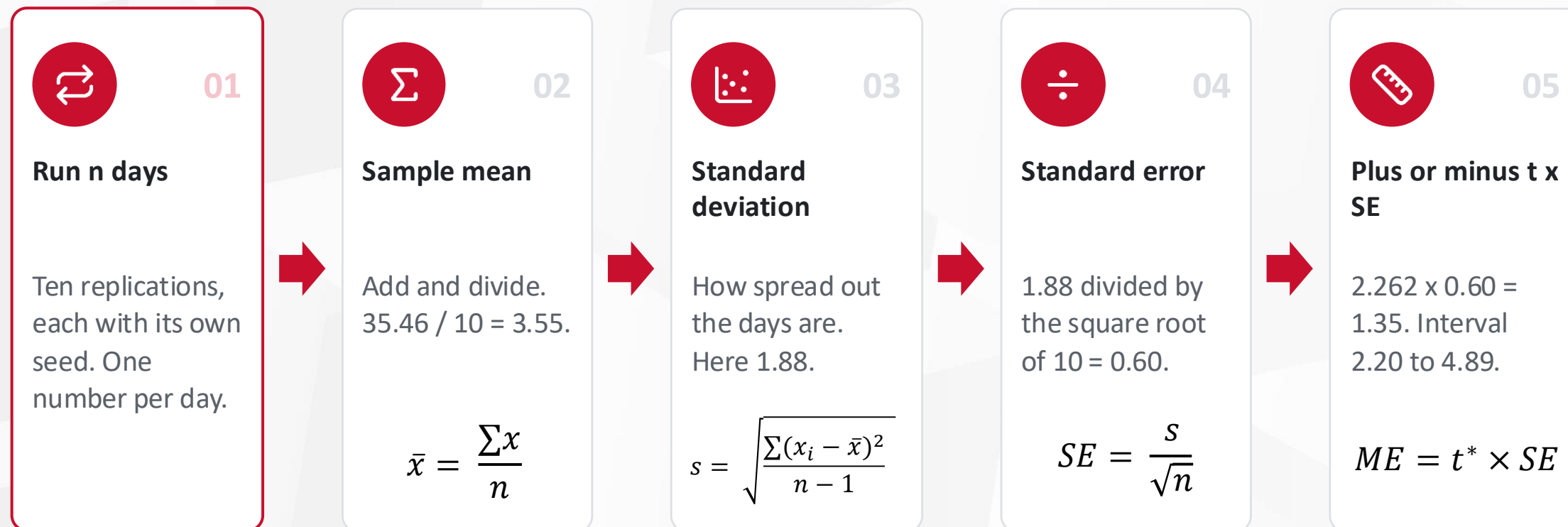
### Half-width (margin of error)

Standard error times  $t = 2.262$ . This is the plus or minus we report.

*Averaging cancels noise. That is why 0.60 is so much smaller than 1.88.*

## SECTION 1: ONE RUN TO AN ANSWER

# Building the interval in five steps



*t comes from a table: 95 percent,  $n - 1 = 9$  degrees of freedom gives 2.262.*

## SECTION 1: ONE RUN TO AN ANSWER

# Reading a confidence interval correctly

- **Confidence interval:** a range built so that, over many studies, 95 percent of such ranges contain the true value.
- Say it like this: the mean wait was 3.55 minutes, 95% CI 2.20 to 4.89.
- A wide interval means you need more replications, not a different story.
- It does NOT say 95 percent of days fall in that range. Days vary much more.
- It only covers random noise. A wrong model gives a precise wrong answer.



## ANALOGY FOR RECALL

## A fishing net

The true average is a fish you cannot see. The interval is a net. Cast this way many times, the net catches the fish 95 times in 100. Any single cast either caught it or did not.

## SECTION 1: ONE RUN TO AN ANSWER

# The interval in Python

```
results = [one_day(seed) for seed in range(1, 11)]

n      = len(results)
mean   = sum(results) / n
sd     = statistics.stdev(results)
se     = sd / math.sqrt(n)

t = 2.262          # 95%, 9 degrees of freedom
half_width = t * se

print(mean - half_width, mean + half_width)
# 2.20  4.89
```

## WHAT THIS CODE DOES | DEMO 13

1. Run the model once per seed. Each run returns one day's average wait.
2. Add the results and divide to get the sample mean.
3. Measure how spread out the ten days are.
4. Shrink that spread by the square root of  $n$  to get the uncertainty of the mean.
5. Multiply by  $t$  from the table to get the plus or minus.

## Section 1: From one run to an honest answer

Q1

Two runs of the same model give 2.1 and 5.8 minutes. Is the model broken? Explain.

Q2

What is the difference between the standard deviation and the standard error of ten replications?

Q3

Five replications give a mean of 4.0 and a standard error of 0.5. Using  $t = 2.776$ , what is the 95 percent interval?

Q4

A student writes: 95 percent of days will have waits between 2.20 and 4.89 minutes. What is wrong?

## SECTION 2

## 02

## How Many Runs, and What Kind of Study

*Decide the question before you press run.*

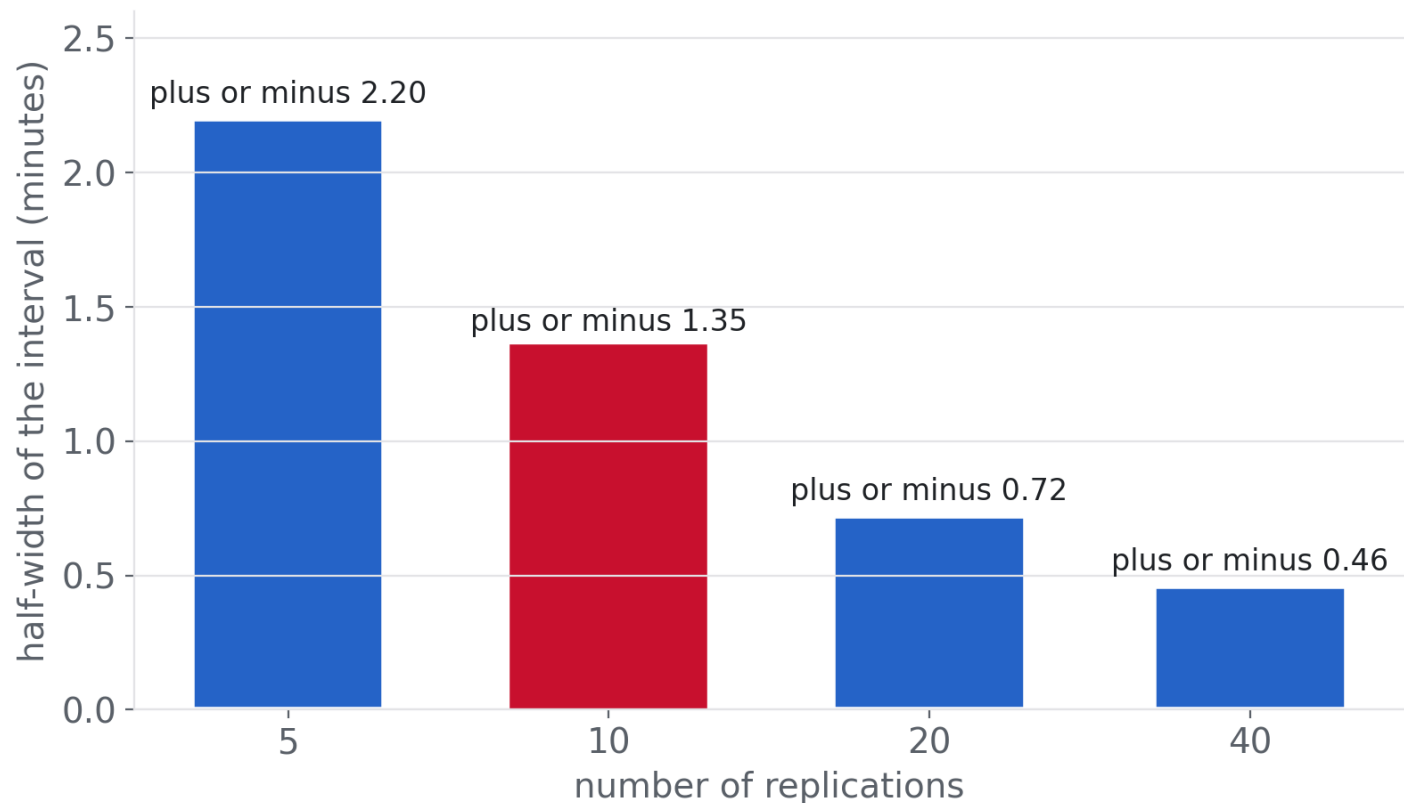
### IN THIS SECTION

- The square-root rule
- Planning replications
- Terminating studies
- Steady-state studies
- Warm-up
- An analysis plan

## SECTION 2: PLANNING THE STUDY

# More runs, narrower interval, but slowly

4 times the runs, only half the uncertainty



### SQUARE-ROOT RULE

**Four times the runs,  
half the uncertainty**

The half-width shrinks with the square root of the number of replications. Going from 10 runs to 40 runs does not make it four times smaller. It halves it.

## SECTION 2: PLANNING THE STUDY

# How many runs do I need?

- Start with a small **pilot study**, such as 10 replications.
- Decide the precision the decision needs, such as plus or minus 0.5 minutes.
- **Runs needed** = runs so far x (half-width now / half-width wanted) squared.
- Run that many, then check the interval you actually got.
- The formula is an estimate. If you miss the target, apply it again.



## QUICK CALCULATION

## Plus or minus 0.5 minutes

Now: 1.35. Wanted: 0.5.  $1.35 / 0.5 = 2.69$  times narrower.  $2.69$  squared =  $7.25$ .  $10 \times 7.25 = 72.5$ , round up to 73 runs.  
Result: 3.60 plus or minus 0.38.  
Target met.

## SECTION 2: PLANNING THE STUDY

# Two kinds of study

## ▶ Terminating study

- Has a natural start and a natural end.
- Example: the 11:00 to 13:00 lunch rush.
- Example: one enrollment day at the registrar.
- The starting condition is real, so keep it.
- Each replication runs from opening to closing.

## ∞ Steady-state study

- Has no natural end. It runs on and on.
- Example: a 24-hour data center or a busy website.
- We want the long-run average behavior.
- The empty start is NOT real, so remove a warm-up.
- Runs must be long enough to settle down.

*The question decides the study type, and the study type decides the run length and the warm-up.*

# Same canteen, two questions, two correct answers

## 3.05

### Lunch rush wait

Terminating: 2 hours, starts empty, 30 replications. Plus or minus 0.95.

## 4.66

### Long-run wait

Steady state: 40-hour runs, first 200 minutes dropped. Plus or minus 0.47.

## 4.50

### Theory's answer

Queueing formula for the long run. It falls inside the steady-state interval.

*The lunch rush is lower because it really does start empty. That is the right answer to that question.*

# Warm-up: only for steady-state studies

- **Initialization bias:** early results reflect an artificial starting state, not normal operation.
- A steady-state model usually starts empty, so early customers wait too little.
- **Warm-up period:** the opening stretch we discard before collecting results.
- Choose it by plotting the average over time and seeing where it levels off.
- Never drop a warm-up in a terminating study. There, the start is the point.



## ANALOGY FOR RECALL

### Baking in a cold oven

You do not time a cake from the moment you switch the oven on. You wait until it reaches temperature. The warm-up period is the oven heating up.

## SECTION 2: PLANNING THE STUDY

# Your analysis plan, before you run anything

| Decision          | Terminating study             | Steady-state study                |
|-------------------|-------------------------------|-----------------------------------|
| Start of each run | The real opening condition    | Any convenient state, often empty |
| End of each run   | The natural closing event     | A long, fixed length you choose   |
| Warm-up           | None                          | Drop it; choose from a plot       |
| Replications      | Several, one per day          | Several long runs                 |
| What you report   | Average over the whole period | Long-run average after warm-up    |

## Section 2: How many runs, and what kind of study

Q1

A pilot of 10 runs gives plus or minus 2.0. How many runs for plus or minus 1.0?

Q2

You simulate a bank branch from 9:00 to 15:00. Terminating or steady-state? Should you drop a warm-up?

Q3

You simulate a web server that never shuts down. Terminating or steady-state? Why does it need a warm-up?

Q4

Why is the lunch-rush wait (3.05) lower than the long-run wait (4.66) for the same canteen?

## SECTION 3

## 03

## Verification: Did We Build It Right?

*Python tells you when code crashes. It never tells you when it is wrong.*

### IN THIS SECTION

- What verification means
- Traces
- Conservation checks
- Sanity checks
- A hidden bug
- Common mistakes

## SECTION 3: VERIFICATION

# What verification means

- **Verification:** checking that the code does what the model design says it should.
- It asks: did we build the model RIGHT?
- A model can run without errors and still be wrong.
- Verification looks for mistakes in the code, the logic, and the bookkeeping.
- It does not ask whether the design matches the real world. That comes next.



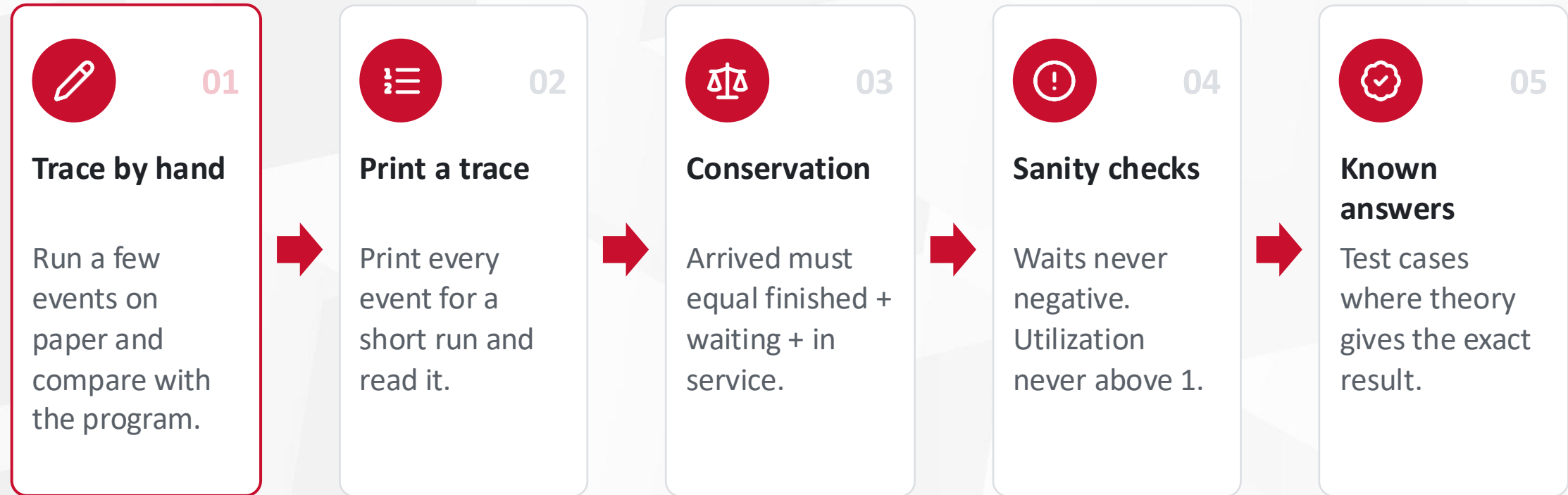
ANALOGY FOR RECALL

## Proofreading a recipe card

Did you copy every step and every amount exactly as the recipe says? Proofreading catches typing mistakes. It cannot tell you whether the recipe itself tastes good.

## SECTION 3: VERIFICATION

# Five ways to verify a model



*Demo 1 from the DES week was a hand trace. Demo 16 adds conservation and sanity checks.*

## SECTION 3: VERIFICATION

# A bug that produced no error message

## 245 = 245

### Correct model

245 arrived. 245 accounted for.  
Conservation check passes.

## 220 of 238

### Buggy model

One leftover test line made  
customers vanish when the line  
reached 3.

## 18

### Customers missing

And the buggy cashier looked  
less busy: 0.69 instead of 0.77.

*Without the check, the buggy model would tell the manager the canteen is fine.*

## SECTION 3: VERIFICATION

# A conservation check in Python

```
arrived      = count["arrived"]
finished     = count["finished"]
waiting      = len(cashier.queue)
at_counter   = cashier.count

accounted = finished + waiting + at_counter

if accounted == arrived:
    print("PASS")
else:
    print("FAIL:", arrived - accounted, "missing")
```

## WHAT THIS CODE DOES | DEMO 16

1. Count every customer the moment they arrive.
2. Count every customer the moment they finish.
3. At the end, also count who is still waiting and who is at the counter.
4. Those three must add up to the arrivals. Every customer is somewhere.
5. If they do not, customers were lost or invented, and the model is wrong.

## SECTION 3: VERIFICATION

# Bugs that simulation students actually make

- Forgetting to release a resource, so the queue grows forever.
- Passing a mean where the function expects a rate, as in expovariate.
- Mixing units: arrival gaps in minutes, service times in hours.
- Using the same seed for every replication, so all runs are identical.
- Counting statistics during the warm-up of a steady-state study.



HABIT TO BUILD

## Check after every change

Keep the checks in your code permanently. Every time you edit the model, run them again. A check you only ran once protects only that one version.

## Section 3: Verification

Q1

In your own words, what question does verification answer?

Q2

At the end of a run: 300 arrived, 280 finished, 12 waiting, 1 at the counter. Does the conservation check pass? What does that tell you?

Q3

Why is a bug that makes the cashier look LESS busy more dangerous than one that makes it look busier?

Q4

Name two sanity checks you could add to any queueing model.

## SECTION 4

## 04

## Validation: Did We Build the Right Thing?

*A perfectly coded model of the wrong system is still wrong.*

### IN THIS SECTION

- What validation means
- Verification vs validation
- Ways to validate
- Comparing with real data
- Calibration
- Careful wording

# Verification and validation are different questions



## Verification

- Did we build the model RIGHT?
- Compares the code with the design.
- Uses traces, conservation, sanity checks.
- Can be done without the real system.
- Recipe analogy: proofreading the card.



## Validation

- Did we build the RIGHT model?
- Compares the model with the real system.
- Uses real data and expert judgment.
- Needs observations of the real system.
- Recipe analogy: tasting the dish.

*A model can pass verification and still fail validation.*

## SECTION 4: VALIDATION

# Ways to validate a model

- **Compare with real data:** do the model's results match observations of the system?
- **Face validity:** do the people who run the system find the behavior believable?
- **Extreme conditions:** with no customers, is the wait zero? With too many, does the line grow?
- **Historical check:** can the model reproduce a past day whose outcome you know?
- Use more than one. Each catches different problems.



## EVERYDAY EXAMPLE

## Trusting a weather app

You trust a weather app because its forecasts have matched the real weather often enough. Validation is how a model earns that same trust.

## SECTION 4: VALIDATION

# Comparing the model with the real canteen

| Source                        | Average wait | 95% interval | Overlaps real data? |
|-------------------------------|--------------|--------------|---------------------|
| Real canteen, 6 days timed    | 3.87 min     | 3.04 to 4.69 | -                   |
| Model, service 1.5 min        | 3.68 min     | 3.13 to 4.24 | Yes: consistent     |
| Model, service guessed at 1.2 | 1.77 min     | 1.52 to 2.03 | No: reject          |

*The six observed days are example numbers for class. In your project they come from your own stopwatch.*

## SECTION 4: VALIDATION

# Calibration, and saying it carefully

- **Calibration:** adjusting uncertain inputs until the model matches observed data.
- Do not validate on the same data you calibrated with. That proves nothing.
- Keep some real observations aside for the validation check.
- Overlapping intervals mean: no evidence the model is wrong.
- They never mean: the model is proven correct.



## ANALOGY FOR RECALL

## Studying with the exam paper

If you memorise the exact exam questions, a perfect score says nothing about what you learned. Validating on your calibration data is the same trick.

## Section 4: Validation

Q1

Classify each as verification or validation: (a) queue length is never negative; (b) average wait matches stopwatch data.

Q2

Your model's interval is 1.5 to 2.0 minutes. Real data gives 3.0 to 4.6. What should you do next?

Q3

Why should you not validate a model using the same data you used to calibrate it?

Q4

A classmate says their model is 'proven correct' because the intervals overlap. How would you reword that?

SECTION 5

# 05

## Sensitivity and Robust Advice

*What if our inputs are a little wrong?*

### IN THIS SECTION

- Sensitivity analysis
- Reading a sensitivity table
- Robust recommendations
- Evidence to report
- Writing the result

## SECTION 5: SENSITIVITY

# What if the service time is not exactly 1.5?

| Service time | Wait with 1 cashier | Wait with 2 cashiers |
|--------------|---------------------|----------------------|
| 1.3 minutes  | 2.20 +/- 0.27 min   | 0.15 +/- 0.02 min    |
| 1.5 minutes  | 3.68 +/- 0.56 min   | 0.23 +/- 0.05 min    |
| 1.7 minutes  | 7.59 +/- 2.03 min   | 0.33 +/- 0.05 min    |

*30 replications of an 8-hour day for each cell.*

# Reading the sensitivity table

- **Sensitivity analysis:** rerunning the study with inputs changed across plausible values.
- Down the 1-cashier column: a 30 percent change in service time more than triples the wait.
- Queues are most sensitive when the server is already busy most of the time.
- Across every row: two cashiers win by a wide margin.
- **Robust recommendation:** advice that stays the same across the plausible inputs.



ANALOGY FOR RECALL

## Stress-testing a bridge

Engineers do not only test a bridge at its expected load. They test heavier loads too. If it still stands, they trust it. A robust recommendation is a bridge that still stands.

## SECTION 5: REPORTING

# The evidence a credible study reports

| Evidence            | What to show                           | From this week                              |
|---------------------|----------------------------------------|---------------------------------------------|
| <b>Precision</b>    | Mean with a 95% interval and n         | 3.55, 95% CI 2.20 to 4.89, n = 10           |
| <b>Study design</b> | Study type, run length, warm-up, seeds | Terminating, 8 h, no warm-up, seeds 1 to 10 |
| <b>Verification</b> | Which checks were run and passed       | Conservation and sanity checks passed       |
| <b>Validation</b>   | Comparison with real data              | Intervals overlap with 6 observed days      |
| <b>Sensitivity</b>  | Does advice survive input changes?     | 2 cashiers best for service 1.3 to 1.7      |

# Writing the result sentence

- Start with the estimate and its interval, and say how many replications.
- Name the study type, the run length, and any warm-up.
- Say which verification and validation checks passed.
- End with the recommendation and how robust it is.
- **Example:** Over 30 replicated 8-hour days, one cashier gave a mean wait of 3.68 minutes (95% CI 3.13 to 4.24). Two cashiers cut this to 0.23 minutes, and remained best for service times from 1.3 to 1.7 minutes.

## Section 5: Sensitivity and robust advice

Q1

What does sensitivity analysis test?

Q2

In the table, the 1-cashier wait more than triples between 1.3 and 1.7 minutes. Why is the queue so sensitive here?

Q3

What makes a recommendation robust?

Q4

List the five kinds of evidence a credible simulation report should contain.

# Terms from this week

| Term                       | Formal meaning                                       | Analogy for recall            |
|----------------------------|------------------------------------------------------|-------------------------------|
| <b>Replication</b>         | One independent run of the model with its own seed   | One visit to the restaurant   |
| <b>Standard error</b>      | Standard deviation divided by the square root of $n$ | How shaky the average is      |
| <b>Confidence interval</b> | Estimate plus or minus $t \times$ standard error     | A net cast for the true value |
| <b>Terminating study</b>   | A study with a natural start and end                 | One lunch rush                |
| <b>Steady-state study</b>  | Long-run behavior of a system with no natural end    | A shop that never closes      |
| <b>Warm-up period</b>      | Opening stretch discarded before collecting results  | Waiting for the oven to heat  |
| <b>Verification</b>        | Checking the code matches the design                 | Proofreading the recipe card  |
| <b>Validation</b>          | Checking the model matches the real system           | Tasting the dish              |

# What to remember from this week



## One run is one day

Random models give random answers. Replicate, then average.



## Report an interval

Mean plus or minus  $t \times$  standard error, with  $n$  stated.



## Square-root rule

Four times the runs halves the uncertainty. Plan with a pilot.



## Know your study type

Terminating keeps its start. Steady-state drops a warm-up.



## Verify, then validate

Right code first, then the right model. They are different tests.



## Test sensitivity

Good advice survives small errors in the inputs.

**Thank you!**